DO NOW — RECALL

Lesson 055 · Choosing Data Structures for Problems

Name: _____ Class: _____ Date: _____

1. What is a hash collision?

2. When can linked-list insertion be O(1)?

3. What is a BST's advantage when balanced?

4. An empty queue receives 4, 8 and 2 in that order. Repeatedly dequeue an item and push it onto an empty stack until the queue is empty. Trace the sequence returned by popping the stack until it is empty.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Two different keys map to the same bucket.

2. Model answer

When a reference to the insertion node is already known.

3. Model answer

Ordered search can require O(log n) comparisons.

4. Model answer

2, 8, 4. The queue supplies 4, 8, 2; the stack reverses that order.

By the end, you can…

• Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

• Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

• Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

Vocabulary — know these terms

operation — Action such as lookup or insertion.

random access — Direct access by numeric index.

trade-off — A benefit weighed against a cost.

complexity — Growth in resource use.

NEW LEARNING

Read, annotate and explain

Discover

Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

Trace

Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

Evaluate

Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose a structure for a browser's Back action and justify.

Explain the trace

Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Match an undo feature, printer queue, pupil ID lookup and road map to a structure. [4]

Question 2

A timetable stores fixed grid of rooms and periods. Recommend a 2D array or graph and explain operations. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate using List<T> versus linked list for a million-item collection, focusing on frequent random index access versus middle insertion with a known node. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The same data structure is always optimal for every collection.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.